

UDY OF PERFECT SOUND ABSORBING POROUS MATERIAL WITH PERIODIC CONICAL HOLE PROFILE

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with plane wave radiation condition. The two microphones transfer function method is used to calculate the reflection and sound absorption coefficients from the surface average acoustic pressures P1 and P2 at planes 1 and 2 positions. The reflection and sound absorption coefficients are given by

$$R = e^{2j k (L_m + s)} (H_{12} - e^{j k s}) / (e^{j k s} - H_{12}), \quad \alpha = 1 - |R|^2 \quad (1)$$

with k the wavenumber and H12 the transfer function, s=30mm and Lm=120 mm. Résultats

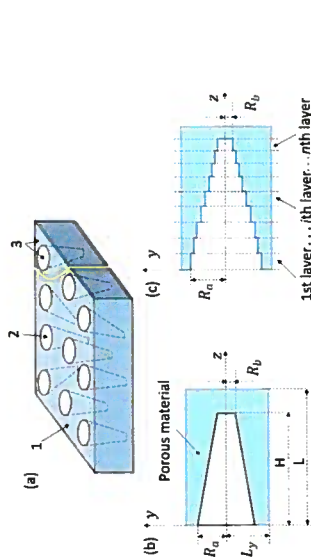


Figure 1: Porous material: (a) with decreasing hole profile (b) conical hole and (c) a series of cylindrical holes.

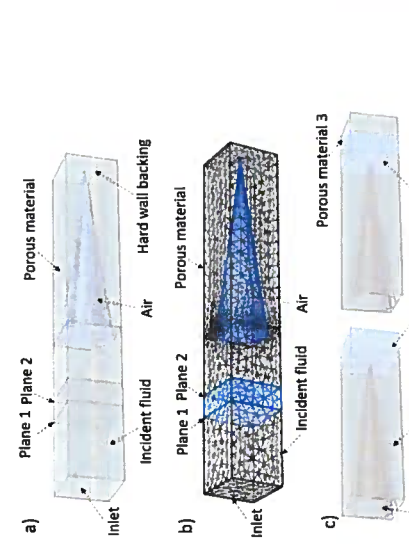


Figure 2: Numerical model: (a) porous layer with conical hole (b)

ous materials used in several ie reduction is limited at low tion can be improved at low thickness, however if the penetration depth, the sound c limit so that any additional vement for the absorption oposed a design of porous ic decreasing hole profiles od (FEM). They showed rmance with large frequency critical depth was overcome. ion of the porous materials ion limit, classical wedge- ot of double porosity or holes Figure 1) can be used. ial design with a periodically esented and its acoustic numerically. Multi-layered al with embedded periodic d with other conventional demonstrated that the sound posed designs is significantly r band.

ons of porous material

terial (1) containing a set of (2). Figure 1(b) presents a th a conical hole, which is lindrical holes of decreasing The thickness of the porous is the length of the conical e are denoted by Ra and Rb-. umerical geometry and mesh sed for the calculations of the

Table 1: Porous material properties.

Materials	ϕ	σ (N s m ⁻²)	α_∞	Λ (μm)	Λ' (μm)
A	0.99	123 170	1	22.25	44.49
B	0.99	17 717	1	58.40	116.8
C	0.99	26 000	1.02	150	300
D	0.99	13 904	1	65.88	131.8

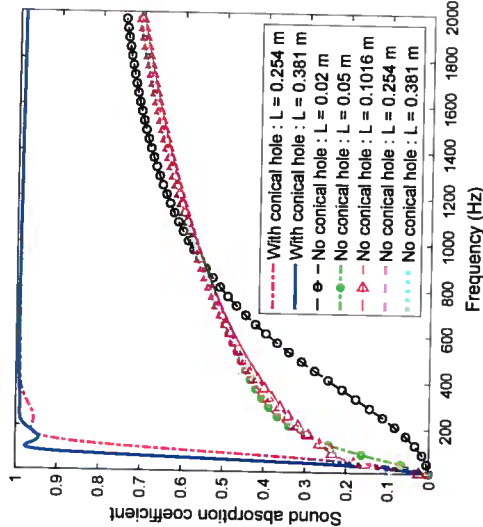


Figure 3: Comparison of the sound absorption coefficient.

The analytical sound absorption coefficient of the porous material with a series of cylindrical holes using the double porosity model is compared with FEM result in Fig. 4. The i^{th} layer is modeled by its transfer matrix T_i [4] and the global matrix T is obtained by multiplying all the matrices of the different layers $T = T_1 T_2 \dots T_n$ [1]. The comparison in Fig. 4 is presented using the porous material D from Table 1 for $n=12$ and $n=20$ material layers with a thickness of 152.4 mm (6 inches), $L_y = 15$ mm, $R_a = 13$ mm, $R_b = 1$ mm; and $H=150$ mm. The analytical results of the sound absorption coefficient agree well with FEM result. The sound absorption presents a large frequency band.

Figure 5 shows the absorption coefficient of porous material A from Table 1 with periodic conical hole combined with one and two other porous layers (see Fig. 2(c)). Porous materials 2 and 3 in Fig. 2(c) have respectively the same properties as B and C in Table 1 with a thickness of 20 mm, $L=0.254$ m, $H=0.25$ m, $L_y=30$ mm, $R_a=28$ mm and $R_b=1$ mm.

The sound absorption coefficient of the multiple layer system containing periodic conical hole in Fig. 5 is significantly improved compared to the multiple layer system without periodic conical hole.

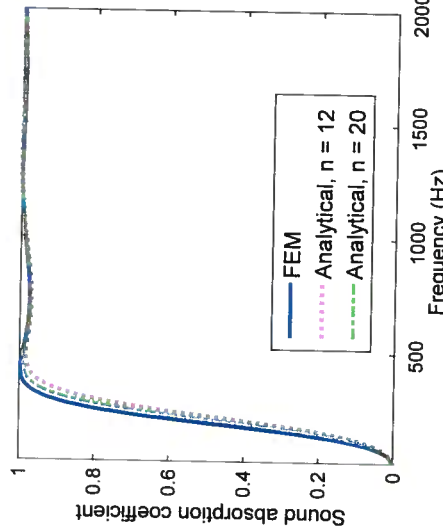


Figure 4: Comparison of FEM with analytical results.

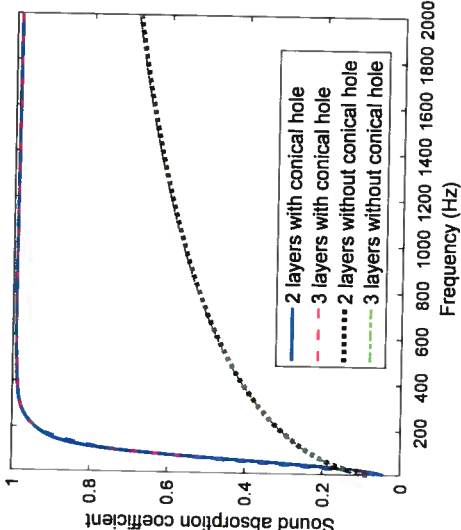


Figure 5: Sound absorption coefficient of multiple layers.

to create multi-layered systems with sound absorption strongly improved compared to classic multi-layered system.

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References

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